

INTRO TO LINEAR TRANSFORMATIONS

Linear Equations in Linear Algebra

INTRODUCTION TO LINEAR TRANSFORMATIONS

- A matrix equation $A\mathbf{x} = \mathbf{b}$ can arise in linear algebra in a way that is not directly connected with linear combinations of vectors. This happens when we think of the matrix A as an object that “acts” on a vector $\mathbf{x}$ by multiplication to produce a new vector called $A\mathbf{x}$.
- For example, the equation

$$\begin{bmatrix} 4 & -3 & 1 & 3 \\ 2 & 0 & 5 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \end{bmatrix}$$

$\uparrow \quad \quad \uparrow \quad \quad \uparrow$
 $A \quad \quad \mathbf{x} \quad \quad \mathbf{b}$

says that that multiplication by A transforms $\mathbf{x}$ into $\mathbf{b}$

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- Solving the equation $Ax = b$ amounts to finding all vectors x in $\mathbb{R}^4$ that are transformed into the vector b in $\mathbb{R}^2$ under the “action” of multiplication by A .
- The correspondence from x to Ax is a function from one set of vectors to another.

INTRODUCTION TO LINEAR TRANSFORMATIONS

- A transformation (or function or mapping) T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that assigns to each vector $\mathbf{x}$ in $\mathbb{R}^n$ a vector $T(\mathbf{x})$ in $\mathbb{R}^m$.
- The set $\mathbb{R}^n$ is called the domain of T , and $\mathbb{R}^m$ is called the codomain of T .
- The notation $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ indicates that the domain of T is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$.

INTRODUCTION TO LINEAR TRANSFORMATIONS

- For $\mathbf{x}$ in $\mathbb{R}^n$, the vector $T(\mathbf{x})$ in $\mathbb{R}^m$ is called the image of $\mathbf{x}$ (under the action of T). The set of all images $T(\mathbf{x})$ is called the range of T .
- The domain of T is $\mathbb{R}^n$ when A has n columns and the codomain of T is $\mathbb{R}^m$ when A has m rows.
- The range of T is the set of all linear combinations of the columns of A , because each image $T(\mathbf{x})$ is of the form $A\mathbf{x}$.
- Mappings associated with matrix multiplication are called matrix transformation and are sometimes denoted by $\mathbf{x} \rightarrow A\mathbf{x}$

MATRIX TRANSFORMATIONS

Example: $A = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix}$, $\mathbf{u} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $\mathbf{b} = \begin{bmatrix} 3 \\ 2 \\ -5 \end{bmatrix}$, $\mathbf{c} = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$

We define a transformation

$T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ by $T(\mathbf{x}) = A\mathbf{x}$,

$$T(\mathbf{x}) = A\mathbf{x} = \begin{bmatrix} 1 & -3 \\ 3 & 5 \\ -1 & 7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} x_1 - 3x_2 \\ 3x_1 + 5x_2 \\ -x_1 + 7x_2 \end{bmatrix}$$

- Find $T(\mathbf{u})$, the image of $\mathbf{u}$ under the transformation T .
- Find an $\mathbf{x}$ in $\mathbb{R}^2$ whose image under T is $\mathbf{b}$.
- Is there more than one $\mathbf{x}$ whose image under T is $\mathbf{b}$?
- Determine if $\mathbf{c}$ is in the range of the transformation T .

LINEAR TRANSFORMATIONS: PROPERTIES

- A transformation (or mapping) T is linear if:
 - (i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for all $\mathbf{u}, \mathbf{v}$ in the domain of T
 - (ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ for all scalars c and all $\mathbf{u}$ in the domain of T .
- Every matrix transformation is a linear transformation.
- Properties (i) and (ii) lead to the following facts:
If T is a linear transformation, then
$$T(\mathbf{0}) = \mathbf{0} \text{ and } T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$
for all vectors $\mathbf{u}, \mathbf{v}$ in the domain of T and all scalars c, d .

SHOWING THAT A TRANSFORMATION IS LINEAR

Example:

- Define $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ by $T(\mathbf{x}) = 3\mathbf{x}$
- Show that T is a linear transformation.

THE MATRIX OF A LINEAR TRANSFORMATION

Linear Equations in Linear Algebra

THE MATRIX OF A LINEAR TRANSFORMATION

- Whenever a linear transformation T arises geometrically or is described in words, we usually want a formula for $T(x)$.
- Every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ is actually a matrix transformation $x \rightarrow Ax$.
- To find a formula for $T(x)$, we can just find A .

THE MATRIX OF A LINEAR TRANSFORMATION

Theorem:

- Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^n.$$

- In fact, A is the $m \times n$ matrix whose j th column is the vector $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j th column of the identity matrix in $\mathbb{R}^n$:

$$A = [T(\mathbf{e}_1) \ \dots \ T(\mathbf{e}_n)]$$

- A is called the standard matrix of the linear transformation T .

THE MATRIX OF A LINEAR TRANSFORMATION

Example:

Find the standard matrix A for the dilation transformation $T(\mathbf{x}) = 3\mathbf{x}$, for $\mathbf{x}$ in $\mathbb{R}^2$.

EXISTENCE AND UNIQUENESS

- A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be onto $\mathbb{R}^m$ if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at least one $\mathbf{x}$ in $\mathbb{R}^n$.
- Equivalently, T is onto $\mathbb{R}^m$ when the range of T is all of the codomain $\mathbb{R}^m$
- “Does T map $\mathbb{R}^n$ onto $\mathbb{R}^m$?” is then an existence question.

EXISTENCE AND UNIQUENESS

- A mapping $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ is said to be one-to-one if each $\mathbf{b}$ in $\mathbb{R}^m$ is the image of at most one $\mathbf{x}$ in $\mathbb{R}^n$.
- Equivalently, T is one-to-one if, for each $\mathbf{b}$ in $\mathbb{R}^m$, the equation $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or none at all.
- “Is T one-to-one?” is a uniqueness question.

EXISTENCE AND UNIQUENESS

Let T be the linear transformation whose standard matrix is

$$A = \begin{bmatrix} 1 & -4 & 8 & 1 \\ 0 & 2 & -1 & 3 \\ 0 & 0 & 0 & 5 \end{bmatrix}$$

- Does T map $\mathbb{R}^4$ onto $\mathbb{R}^3$? Is T a one-to-one mapping?

EXISTENCE AND UNIQUENESS

Theorem:

- Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation. Then T is one-to-one if and only if the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution.

Proof:

We have to prove two statements:

- If T is one-to-one then the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution
- If the equation $T(\mathbf{x}) = \mathbf{0}$ has only the trivial solution then T is one-to-one.

EXISTENCE AND UNIQUENESS

Theorem:

Let $T: \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear transformation and let A be the standard matrix for T . Then:

- A) T maps $\mathbb{R}^n$ onto $\mathbb{R}^m$ if and only if the columns of A span $\mathbb{R}^m$
- B) T is one-to-one if and only if the columns of A are linearly independent.